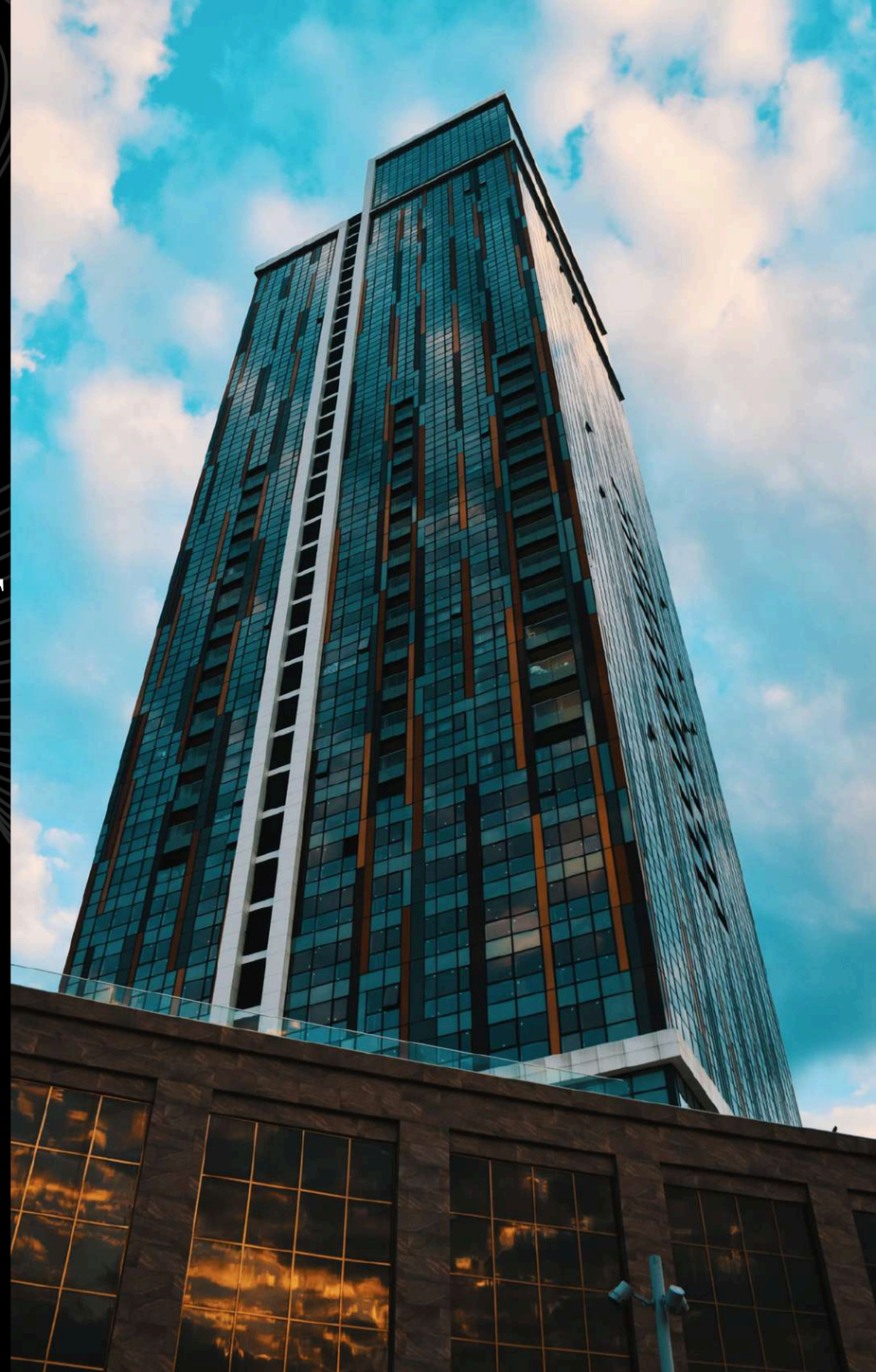
UNDERSTANDING NYC GOVERNMENT JOB TRENDS THROUGH DATA AND NETWORKS



AGENDA

- Introduction And Problem Statement
- Dataset Overview
- Descriptive Statistics
- Data Cleaning Criteria
- Data Observations
- Methodology - Visualization Tools
- Methodology - Social Network Analysis
- Results – Frequency Trends
- Results – Clustering
- Results – Network Analysis
- Conclusions
- References & Dataset Source



INTRODUCTION

- NYC government job postings span numerous agencies and career levels.
- These listings reflect trends in public sector hiring and workforce priorities.
- The dataset reveals opportunities to explore job demand, salary patterns, and inter-agency dynamics.
- This project analyzes the data using visualizations and social network analysis.
- Goal: uncover trends, clusters, and hidden relationships within NYC's public employment landscape.



PROBLEM STATEMENT

- Large number of NYC government job postings, but limited understanding of underlying patterns
- Unclear which agencies frequently collaborate or share similar job roles
- Traditional charts don’t capture structural relationships across departments
- Social network analysis can reveal:
- Inter-agency connections through shared job categories
- Central agencies in the hiring network
- Clusters of similar roles across different departments
- Goal: use network mapping to support smarter job targeting and strategic planning

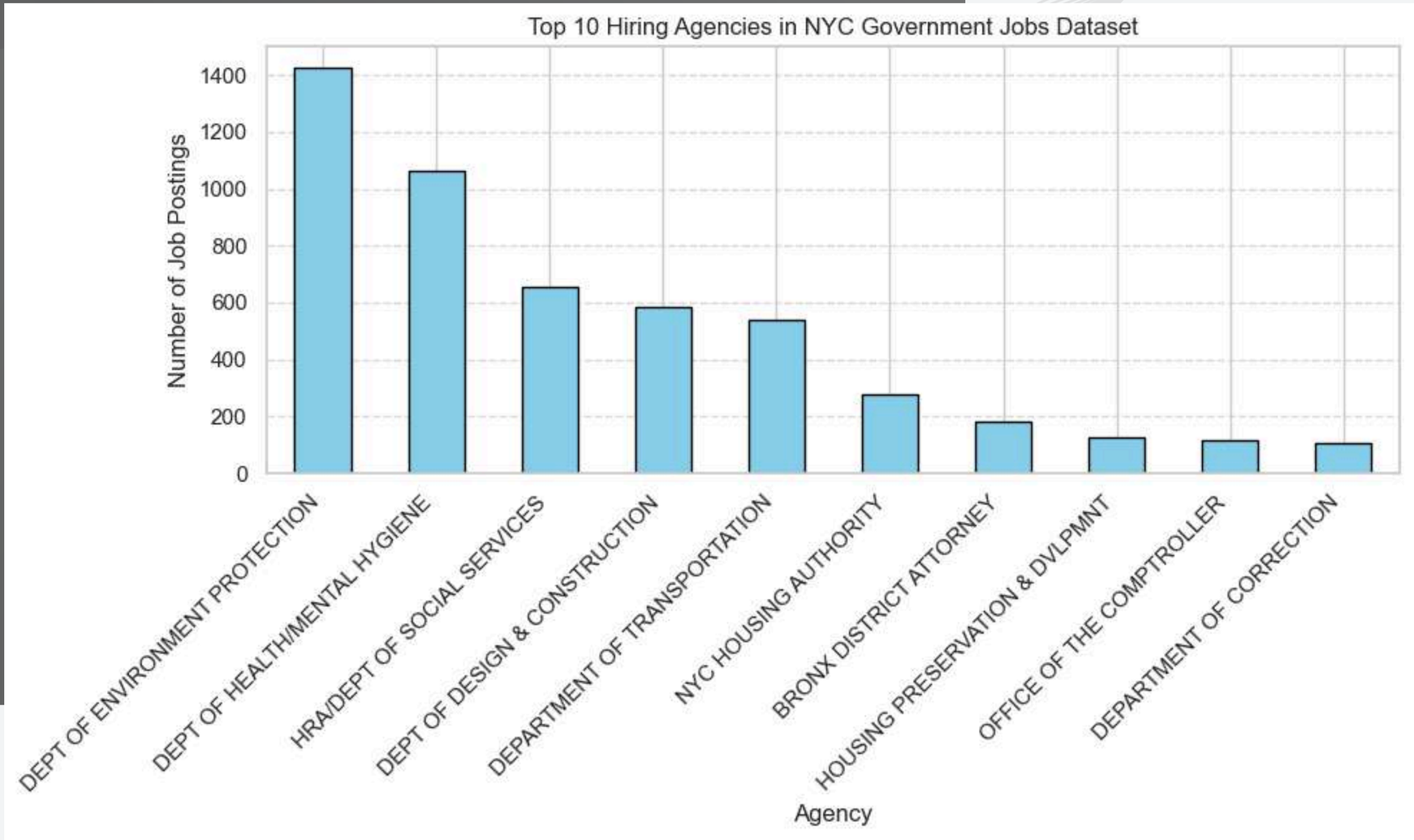
DATASETS

Job ID	Agency	Posting Type	# Of Positions	Business Title	Civil Service Title	Title Classification	Title Code No	Level	Job Category
700602	HOUSING PRESERVATION & DVLPMNT	External	1	Attorney	AGENCY ATTORNEY	Non-Competitive-5	30087	03	Legal Affairs
702651	DEPT OF HEALTH/MENTAL HYGIENE	External	1	Lead Administrator on Duty	FORENSIC MORTUARY TECHNICIAN	Competitive-1	52040	03	Health
560439	DEPT OF ENVIRONMENT PROTECTION	External	1	Associate Staff Analyst	ASSOCIATE STAFF ANALYST	Competitive-1	12627	00	Finance, Accounting, & Procurement
702183	DEPT OF HEALTH/MENTAL HYGIENE	Internal	1	AOT Quality Improvement Specialist, Bureau of Mental Health	CONSULTANT (PUBLIC HEALTH-SOCI	Competitive-1	51613	00	Health
701689	DEPT OF DESIGN & CONSTRUCTION	External	1	Project Manager	ASSISTANT ARCHITECT	Competitive-1	21210	00	Engineering, Architecture, & Planning
681027	DEPT OF HEALTH/MENTAL HYGIENE	Internal	1	Assistant Agency Chief Contracting Officer of M/WBE	ADMINISTRATIVE CONTRACT SPECIA	Competitive-1	1009A	00	Health
583088	DEPT OF ENVIRONMENT PROTECTION	Internal	1	ASSOC. CHEMIST	ASSOCIATE CHEMIST	Competitive-1	21822	01	Public Safety, Inspections, & Enforceme
695252	HRA/DEPT OF SOCIAL SERVICES	External	1	OPERATOR	ASSISTANT PRINTING PRESS OPERA	Competitive-1	92122	00	Building Operations & Maintenance
635845	DEPT OF HEALTH/MENTAL HYGIENE	Internal	1	Data Analyst, Bureau of Tuberculosis Control	CITY RESEARCH SCIENTIST	Non-Competitive-5	21744	01	Constituent Services & Community Pro
550935	DEPT OF ENVIRONMENT PROTECTION	Internal	1	I/C Facilities Plans Reviewer	CIVIL ENGINEER	Competitive-1	20215	03	Engineering, Architecture, & Planning
644723	LAW DEPARTMENT	External	1	Payroll Clerk	CLERICAL ASSOCIATE	Competitive-1	10251	03	Administration & Human Resources Fin
680900	DEPT OF HEALTH/MENTAL HYGIENE	Internal	1	Policy Manager, Bureau of Equitable Health Systems	CITY RESEARCH SCIENTIST	Non-Competitive-5	21744	03	Health Policy, Research & Analysis
638912	DEPT OF DESIGN & CONSTRUCTION	External	1	Deputy Director	ADM CONSTRUCTION PRJ MGR-NM	Competitive-1	8299A	00	Engineering, Architecture, & Planning
681349	DEPT OF HEALTH/MENTAL HYGIENE	External	1	Director of Healthcare Coalition Operations, Bureau of Healthcare and Co	PUBLIC HEALTH EMERGENCY PREPAR	Non-Competitive-5	06853	02	Health
699025	BRONX DISTRICT ATTORNEY	External	1	Supervising Assistant District Attorney- Child Abuse/Sex Crimes Bureau	ASSISTANT DISTRICT ATTORNEY (S	Exempt-4	30114	00	Legal Affairs
649450	TAXI & LIMOUSINE COMMISSION	Internal	1	Data Engineer	CITY RESEARCH SCIENTIST	Non-Competitive-5	21744	03	Constituent Services & Community Pro
703189	DEPT OF ENVIRONMENT PROTECTION	Internal	1	2025-BWT-011-Industrial Resource Management and Permitting Intern	SUMMER COLLEGE INTERN	Non-Competitive-5	10234	00	Engineering, Architecture, & Planning Po
561622	DEPT OF ENVIRONMENT PROTECTION	Internal	1	ACCOUNTABLE MANAGER	CIVIL ENGINEER	Competitive-1	20215	02	Engineering, Architecture, & Planning Po
650672	HRA/DEPT OF SOCIAL SERVICES	Internal	1	ASSISTANT MANAGER	PRINCIPAL ADMINISTRATIVE ASSOC	Competitive-1	10124	03	Social Services
622527	DEPT OF ENVIRONMENT PROTECTION	Internal	1	Division Chief, Contractor Maintenance	ADMINISTRATIVE ENGINEER	Competitive-1	10015	M3	Engineering, Architecture, & Planning
642021	DEPT OF ENVIRONMENT PROTECTION	Internal	1	ASSISTANT PROJECT MANAGER	CHEMICAL ENGINEERING INTERN	Competitive-1	20503	00	Engineering, Architecture, & Planning Po
552763	DEPT OF ENVIRONMENT PROTECTION	Internal	1	Asset Management Program Planner	ASSOCIATE PROJECT MANAGER	Competitive-1	22427	02	Engineering, Architecture, & Planning
699548	HUMAN RIGHTS COMMISSION	External	1	Strategic Initiatives Manager	EXECUTIVE PROGRAM SPECIALIST	Exempt-4	10264	M1	Communications & Intergovernmental A
635394	DEPT OF DESIGN & CONSTRUCTION	External	1	Program Executive	ADMIN ARCHITECT (NON MGRL) M-1	Competitive-1	1000A	00	Engineering, Architecture, & Planning
703530	DEPT OF PARKS & RECREATION	External	1	Supervisor of Payroll Analytical Unit	PRINCIPAL ADMINISTRATIVE ASSOC	Competitive-1	10124	03	Administration & Human Resources
624159	DEPARTMENT OF CORRECTION	External	1	Associate Commissioner, Operations	ASSOCIATE COMMISSIONER (DOC)	Non-Competitive-5	95042	M4	Public Safety, Inspections, & Enforceme
545210	HRA/DEPT OF SOCIAL SERVICES	Internal	1	HEADCOUNT & BUDGET ANALYST	ASSOCIATE STAFF ANALYST	Competitive-1	12627	00	Administration & Human Resources Fin
534939	DEPT OF ENVIRONMENT PROTECTION	Internal	1	Capital Projects/On-Call Design Manager	ADM ENGINEER (NON MGRL)	Competitive-1	1001A	00	Engineering, Architecture, & Planning
698204	CONSUMER AND WORKER PROTECTION	Internal	1	Payment Analyst - 698204	CLERICAL ASSOCIATE	Competitive-1	10251	04	Finance, Accounting, & Procurement
697856	CONSUMER AND WORKER PROTECTION	External	1	Customer Service Representative - 697856	CLERICAL ASSOCIATE	Competitive-1	10251	04	Administration & Human Resources
623022	DEPT OF ENVIRONMENT PROTECTION	External	1	Industrial/Commercial Program Section Lead	CIVIL ENGINEER	Competitive-1	20215	02	Green Jobs Engineering, Architecture, &
694022	DEPT OF DESIGN & CONSTRUCTION	External	1	Safety Auditor	INDUSTRIAL HYGIENIST	Competitive-1	31305	01	Engineering, Architecture, & Planning Po
527822	NYC HOUSING AUTHORITY	Internal	1	Supervisor Electrician	SUPERVISOR ELECTRICIAN	Competitive-1	91769	00	Building Operations & Maintenance
631995	DEPT OF HEALTH/MENTAL HYGIENE	External	1	Case Monitor, Bureau of Mental Health/Assisted Outpatient Treatment	SOCIAL WORKER	Competitive-1	52613	00	Health Social Services
702797	OFFICE OF MANAGEMENT & BUDGET	External	1	Analyst Compliance & Monitoring	BUDGET ANALYST (OMB)	Pending Classification-2	06088	01	Finance, Accounting, & Procurement Po
640136	DEPT OF DESIGN & CONSTRUCTION	External	1	Assistant Commissioner	ADMINISTRATIVE ENGINEER	Competitive-1	10015	M4	Engineering, Architecture, & Planning
639237	DEPT OF HEALTH/MENTAL HYGIENE	External	1	Communications Manager, Bureau of Equity	ADM MANAGER-NON-MGRL FRM M1/M2	Competitive-1	1002C	02	Communications & Intergovernmental A
647907	DEPT OF HEALTH/MENTAL HYGIENE	External	1	Exterminator, Bureau of Veterinary and Pest Control Services	EXTERMINATOR	Competitive-1	90510	02	Health Public Safety, Inspections, & Enf
531327	NYC HOUSING AUTHORITY	External	1	HOUSING ASSISTANT	HOUSING ASSISTANT (HA)	Competitive-1	80201	00	Administration & Human Resources Soc
684379	OFFICE OF MANAGEMENT & BUDGET	Internal	1	Administrative Assistant EXECUTIVE	STATISTICAL SECRETARY (OMB)	Pending Classification-2	05363	01	Administration & Human Resources
693129	FINANCIAL INFO SVCS AGENCY	External	1	WebSphere Administrator	SENIOR IT ARCHITECT	Non-Competitive-5	95711	00	Technology, Data & Innovation

DATASET OVERVIEW

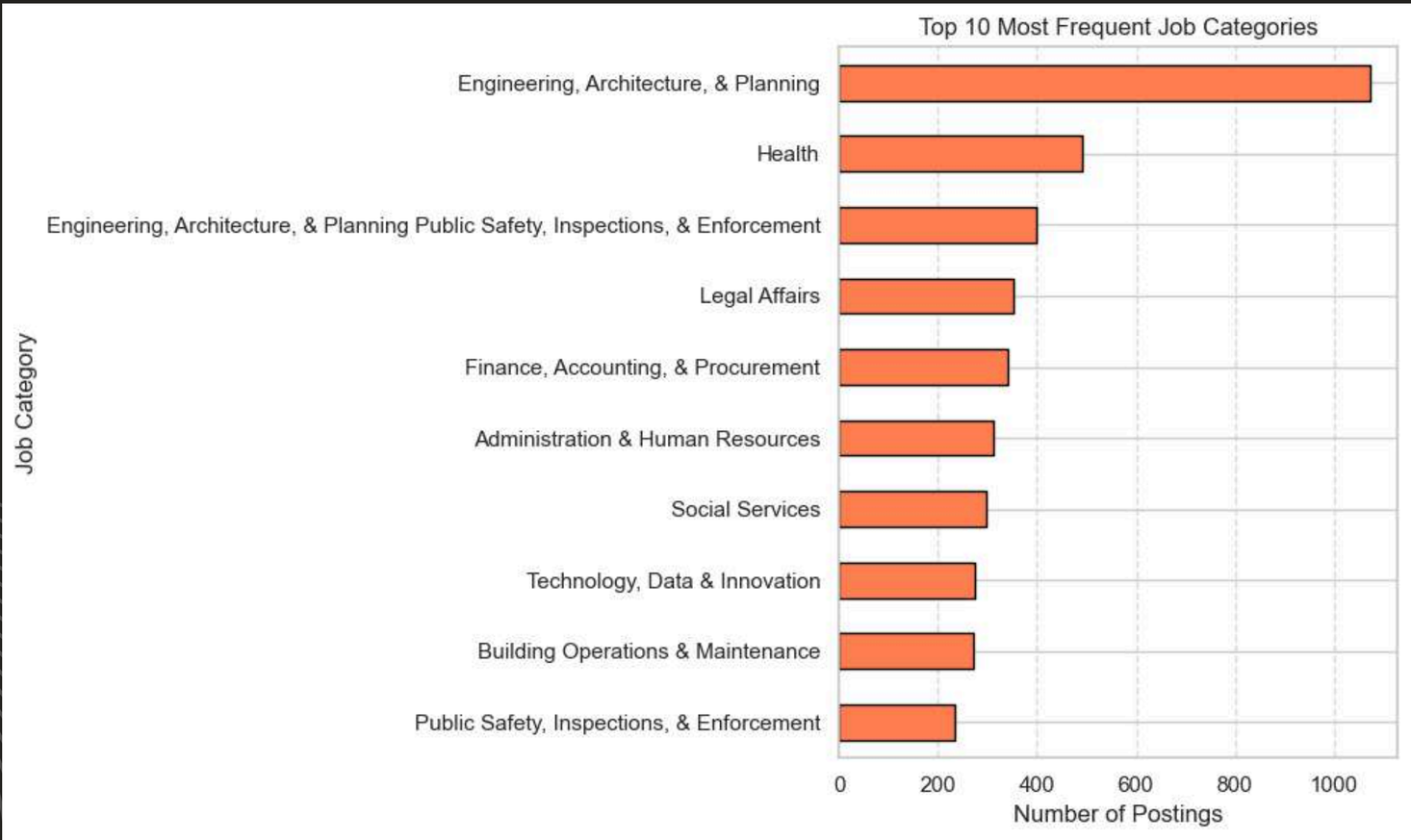
Total Records in the dataset are 6,316 unique job postings, Features 27 (Columns)
Time Period: Ongoing postings, current snapshot at time of extraction
Focus Areas for Network Analysis:

- **Agency**
- **Job Category**
- **Career Level**
- **Full-Time/Part-Time Status**
- **Salary Range**
- **Posting Date**

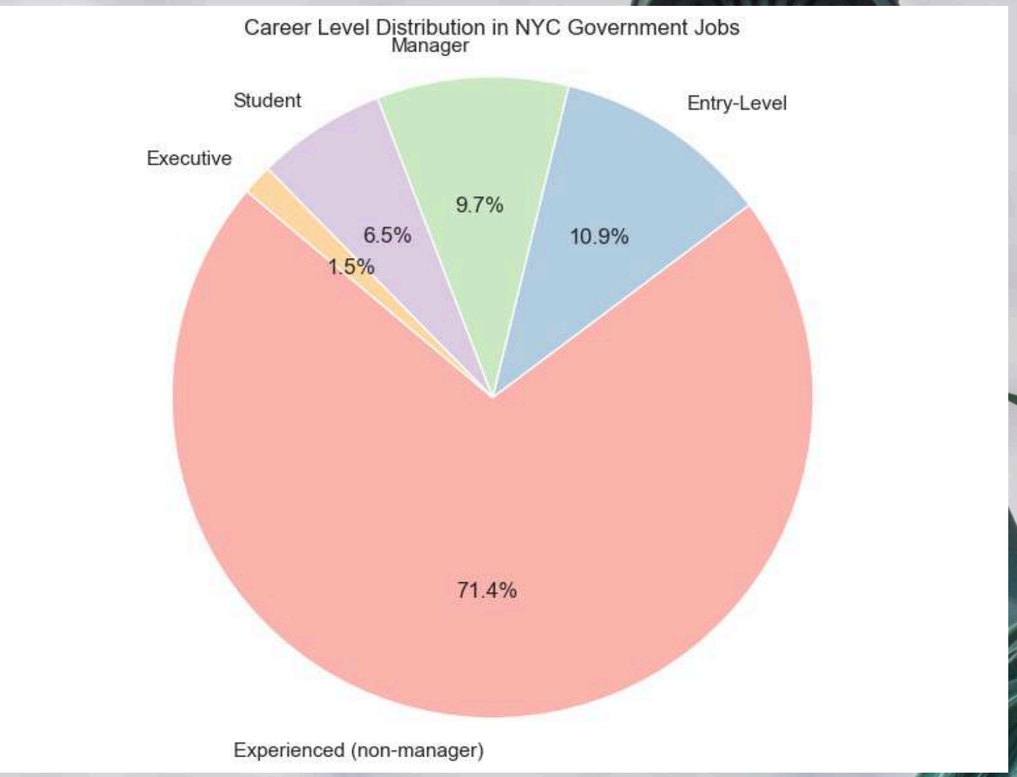


Descriptive Statistics

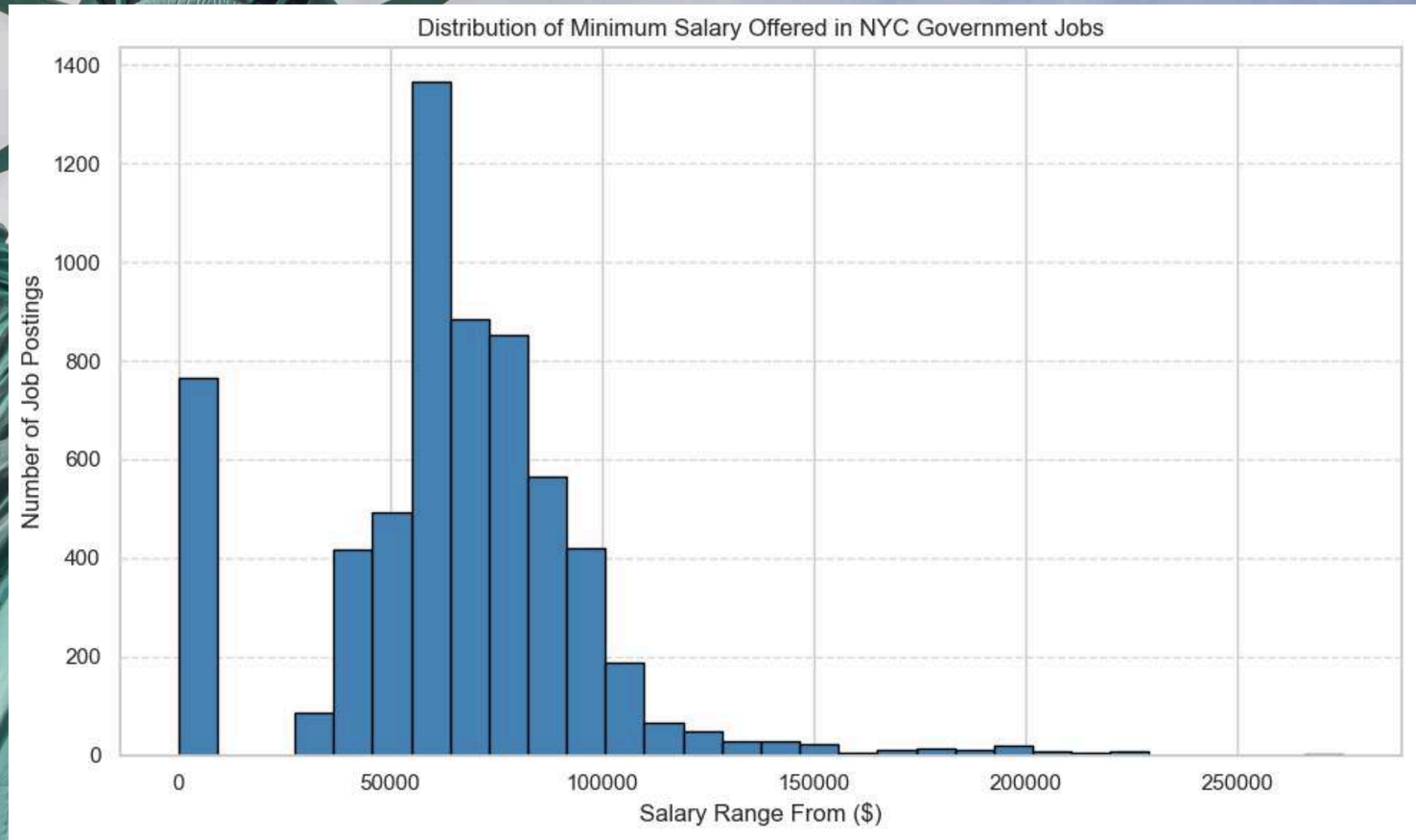
- Key Statistics:
- Total Job Postings: 6,316
- Number of Unique Agencies: ~80
- Top Hiring Agencies: Department of Education, Department of Health, NYC Housing Authority
- Most Common Job Categories: Clerical, Health Services, Administrative, Engineering
- Career Levels:
- Entry-Level → ~60%
- Experienced → ~30%
- Managerial → ~10%
- Salary Range (Minimum):
- Median: ~$55,000
- Range: $15,000 to $200,000+



DATA EXPLORATORY(FOCUSING PROBLEM)



- The chart visualizes the proportion of job postings across different career levels.
- Experienced (non-manager) roles dominate the dataset at 71.4%

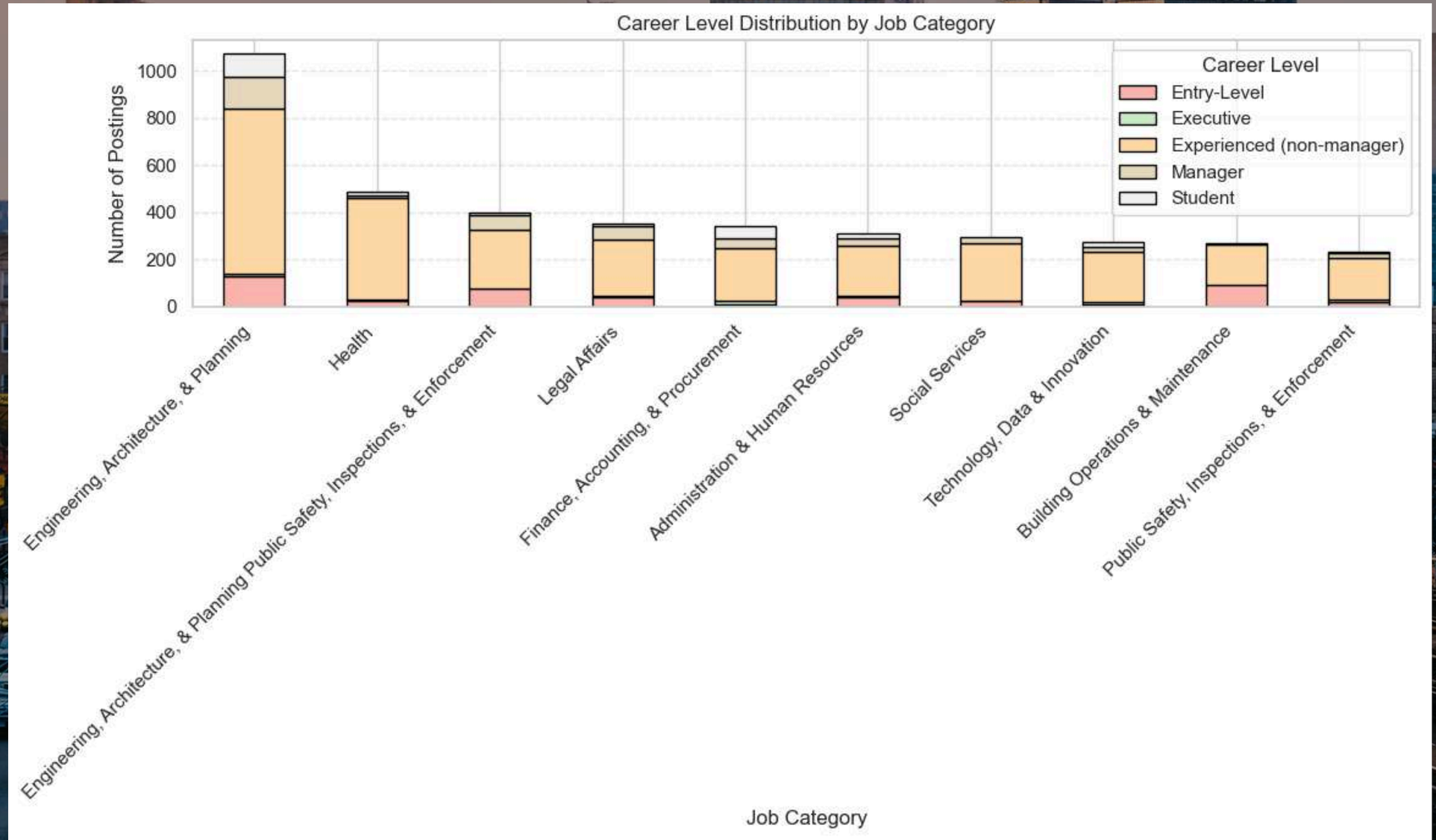


- Most job postings offer starting salaries between $40,000 and $80,000, with a peak around $55,000–$60,000.
- A significant spike near $0–$10,000, which may indicate:

1. Data entry errors
2. Internship/unpaid roles
3. Placeholder values

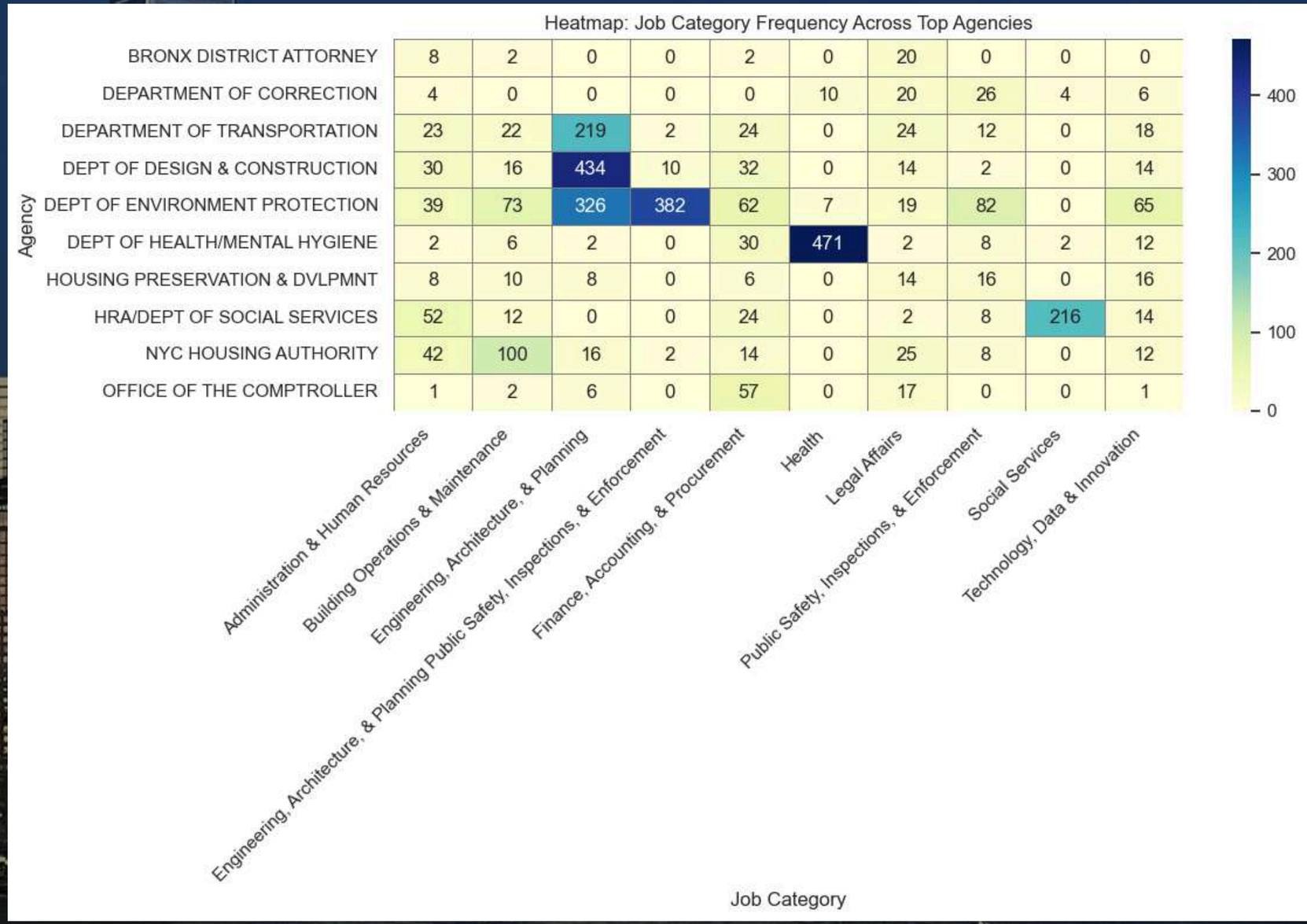
DATA EXPLORATORY

Most NYC government job categories are dominated by experienced (non-manager) roles, with minimal entry-level and executive opportunities, highlighting a mid-level hiring focus.



DATA EXPLORATORY

- Each cell shows the number of postings for a specific job category by an agency.
- Darker cells = more job postings → stronger specialization in that category.
- HRA/Dept of Social Services specializes in Social Services (216 postings).



DATA CLEANING CRITERIA

Dropped incomplete rows with missing values in:

- Agency
- Job Category
- Career Level
- Salary Range From

USING PYTHON

Removed invalid salary data:

- Excluded rows where Salary Range From was 0 or negative.

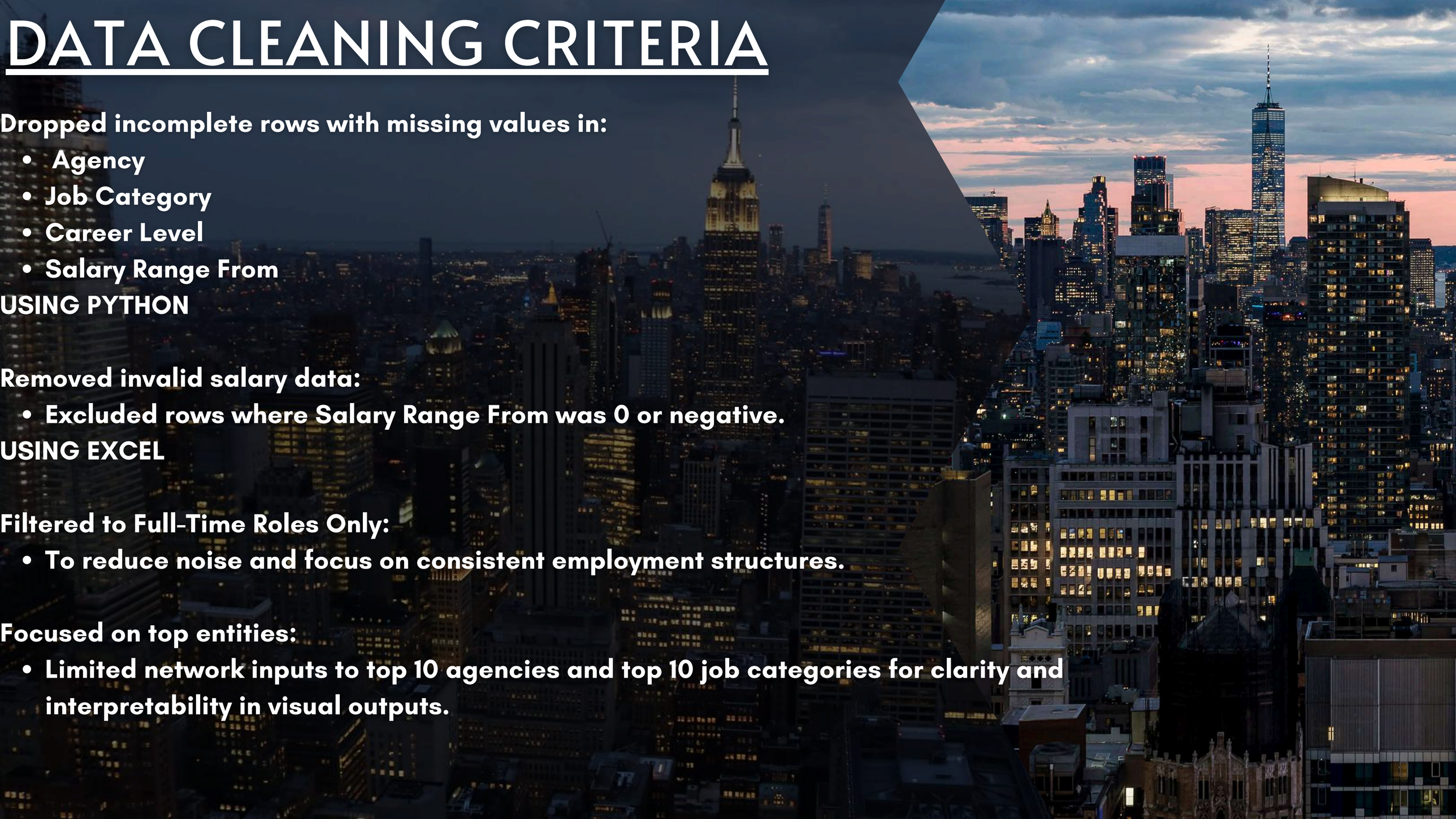
USING EXCEL

Filtered to Full-Time Roles Only:

- To reduce noise and focus on consistent employment structures.

Focused on top entities:

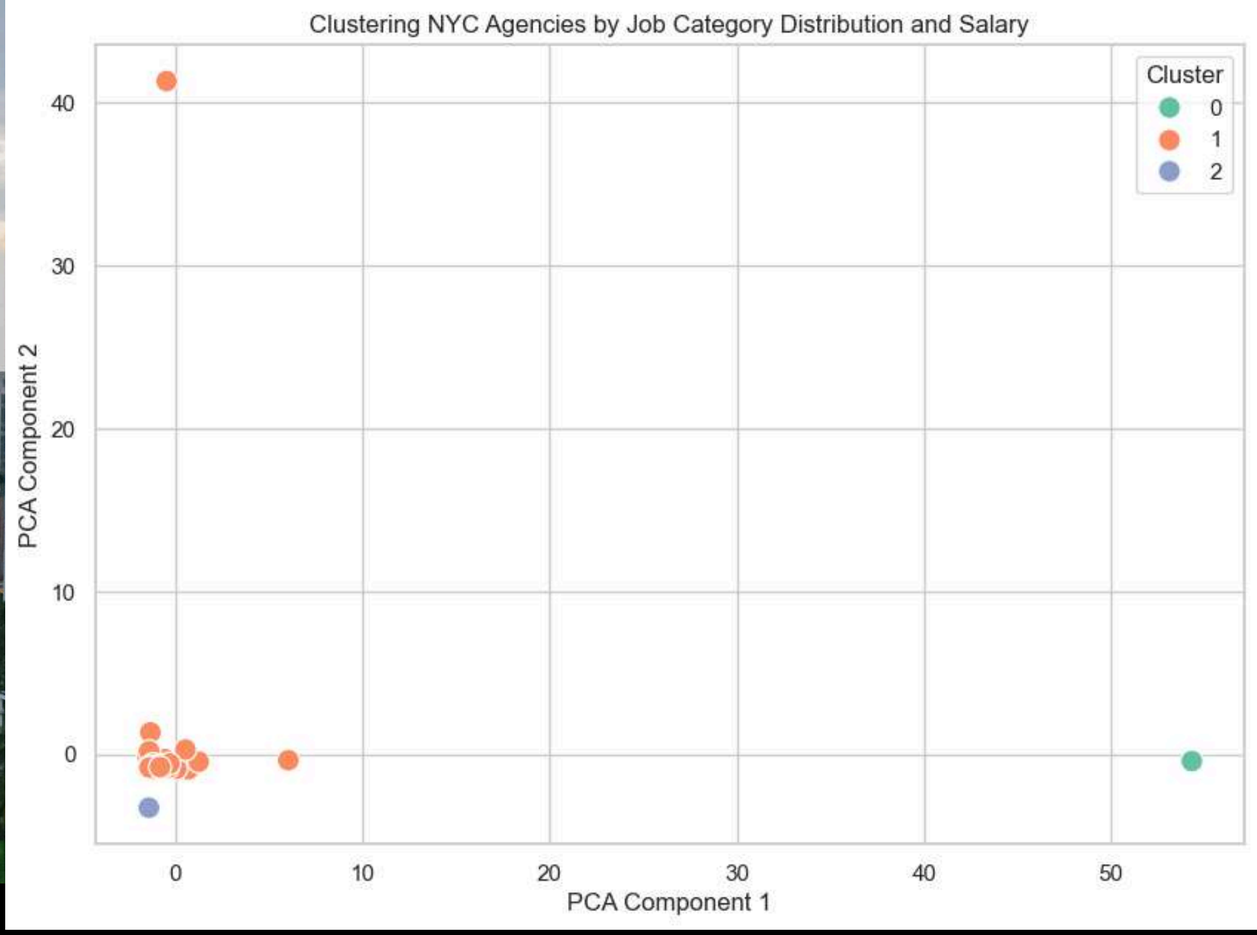
- Limited network inputs to top 10 agencies and top 10 job categories for clarity and interpretability in visual outputs.



DATA OBSERVATION

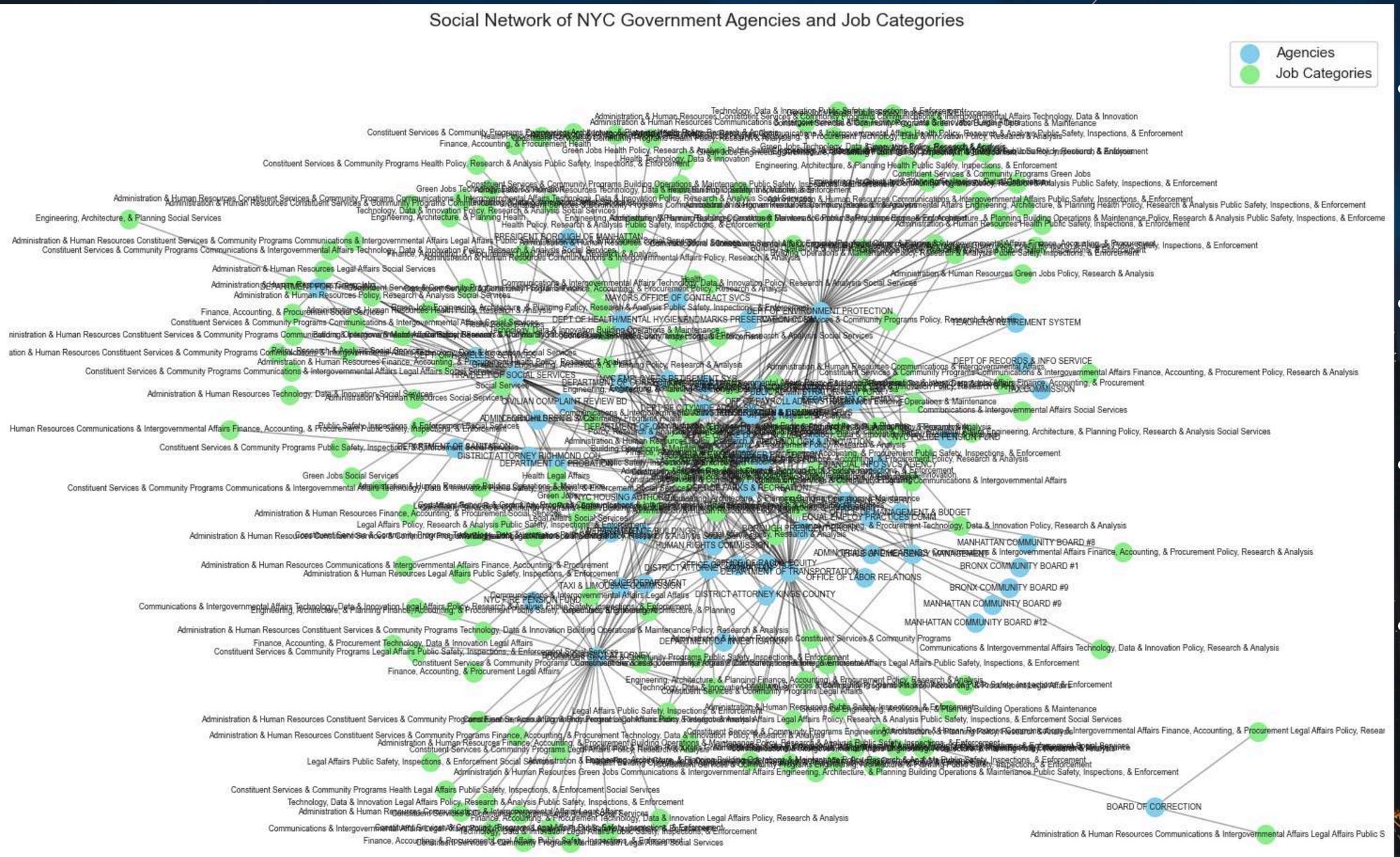
Agency	Avg Salary	Cluster
DEPT OF ENVIRONMENT PROTECTION	61476.598921	0
ADMIN FOR CHILDREN'S SVCS	72554.835165	1
FIRE DEPARTMENT	73611.320000	1
HOUSING PRESERVATION & DVLPMNT	75943.968750	1
HRA/DEPT OF SOCIAL SERVICES	66786.788432	1
...	...	...
DEPT. OF HOMELESS SERVICES	58219.685833	1
DISTRICT ATTORNEY KINGS COUNTY	55046.666667	1
DEPARTMENT OF FINANCE	31366.232432	1
TECHNOLOGY & INNOVATION	86499.258065	1
BRONX DISTRICT ATTORNEY	87006.005587	2

[61 rows x 2 columns]



- Some clusters show specialized agencies (focused on 1–2 roles).
- Others show diversified agencies (broad job category spread).
- Clusters also reveal which agencies offer higher or lower salaries.
- Helps understand each agency’s hiring scale, role type, and workforce strategy.

DATA VISUALIZATION



- Central (blue) agencies are diversified — they hire for many job categories
- Outer agencies are specialized — focused on 1–2 roles
- Agencies with shared job categories have functional similarity
- The diagram shows few connections to executive roles, confirming a mid-level hiring focus.

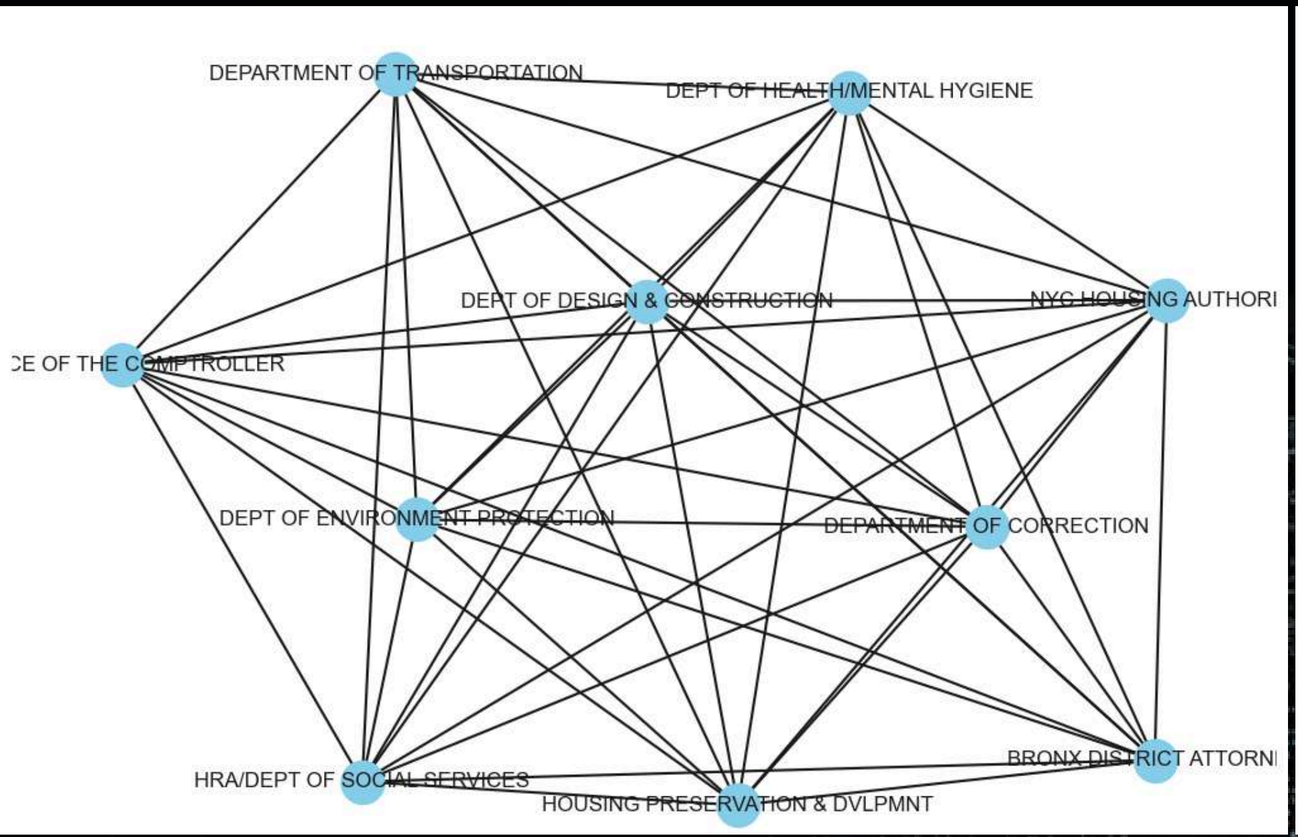
DATA VISUALIZATION



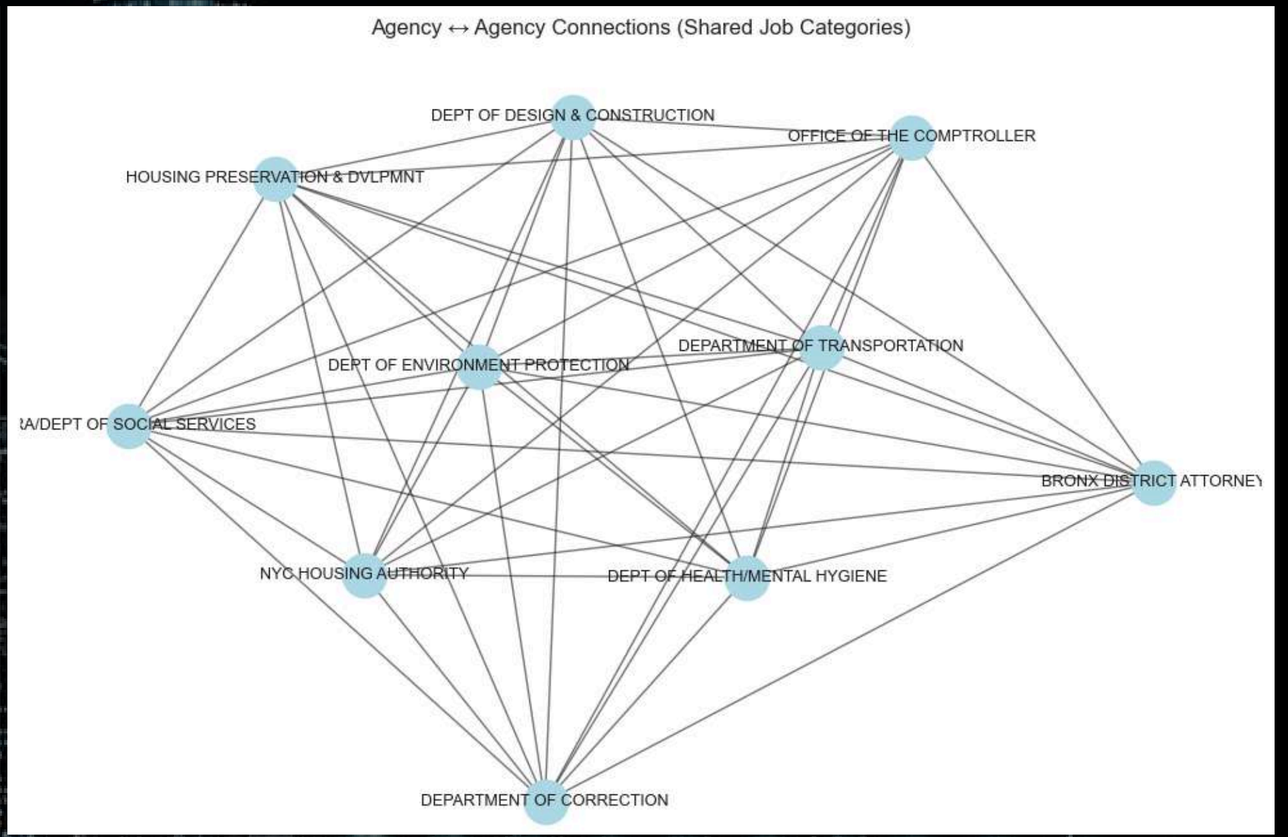
- AGENCIES ARE GROUPED BASED ON SIMILAR JOB CATEGORIES, SHOWING HIRING BEHAVIOR PATTERNS.
- SOME AGENCIES LIKE NYC HOUSING AUTHORITY ARE DIVERSIFIED, HIRING ACROSS MANY ROLES.

DATA VISUALIZATION

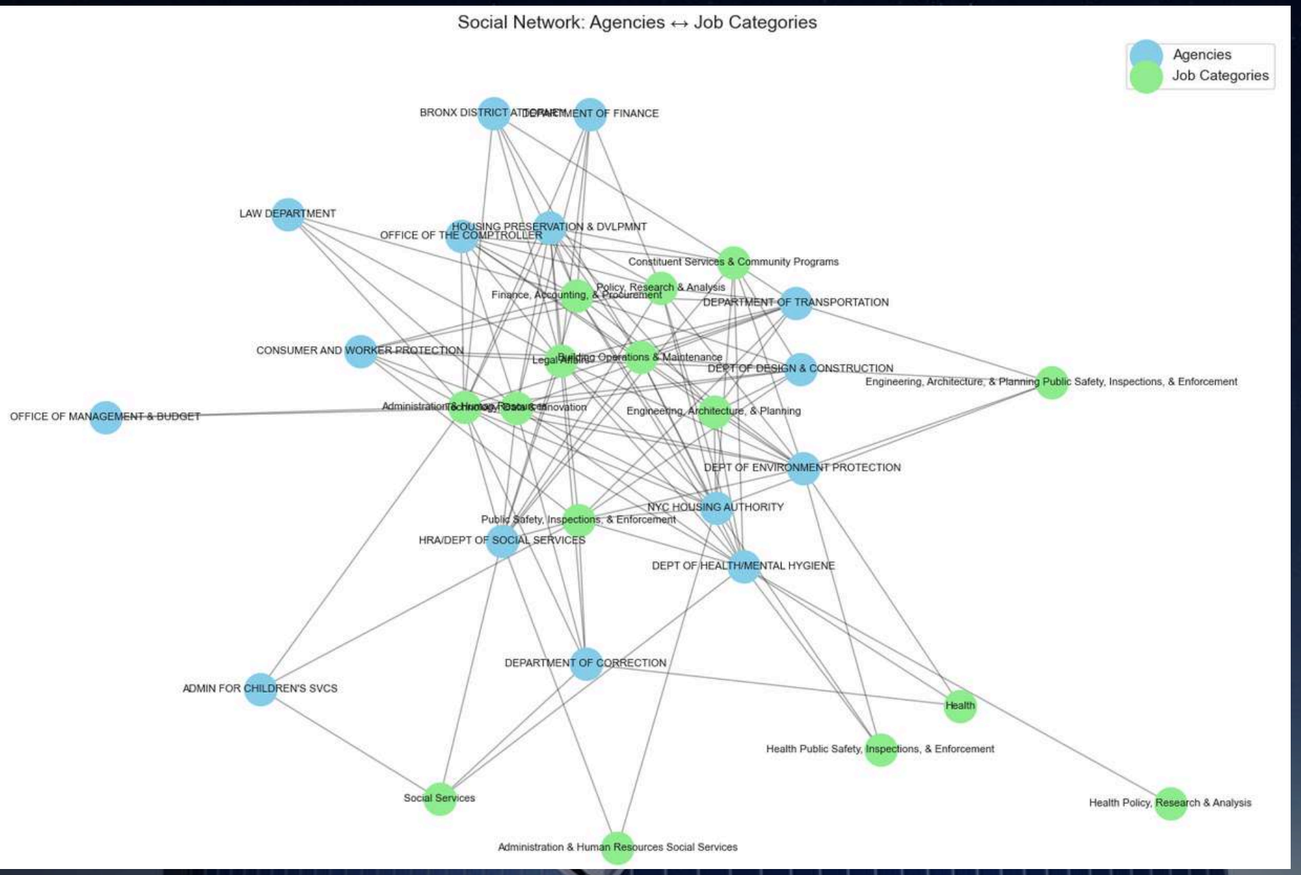
Agency ↔ Agency Projection based on shared Career Levels



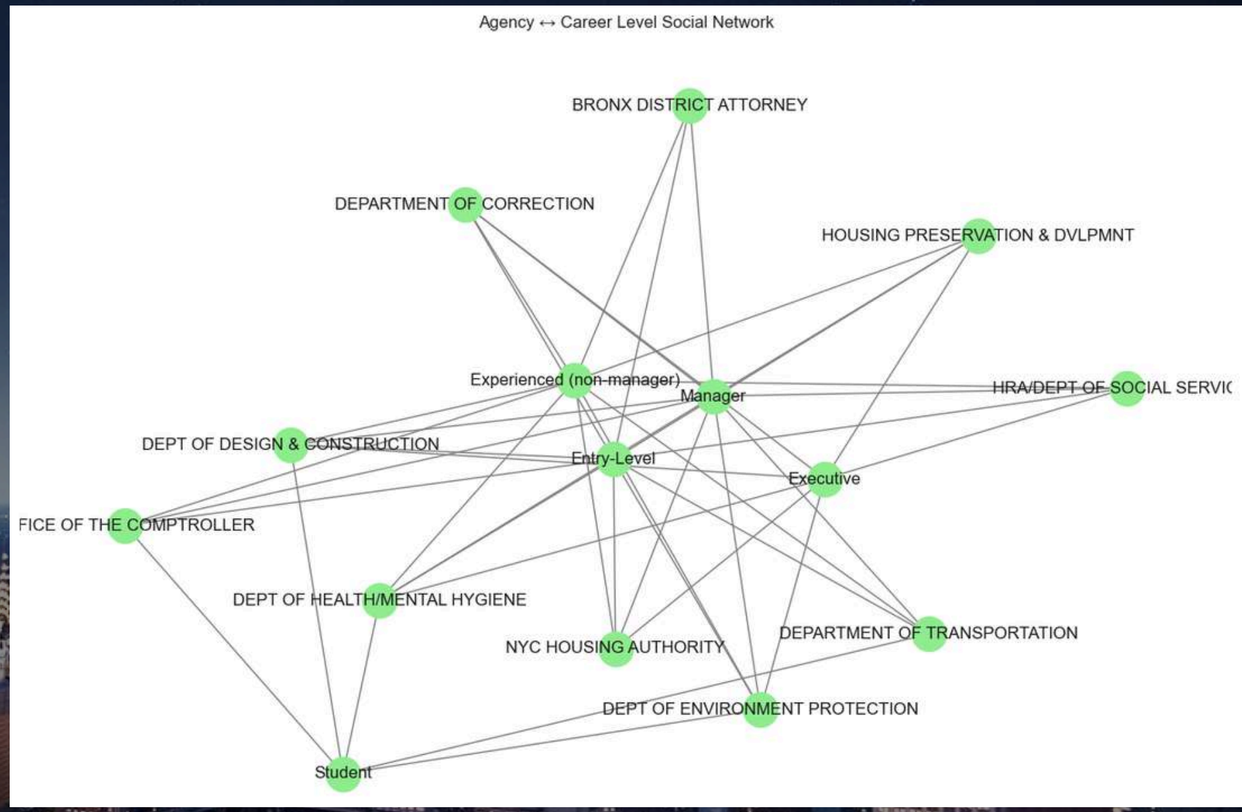
Agency ↔ Agency Connectios(Shared jobs)



DATA VISUALIZATION(CONCLUSION).



- SHOWS WHICH AGENCIES SHARE SIMILAR JOB CATEGORIES (FUNCTIONAL SIMILARITY).
- DIVERSIFIED AGENCIES APPEAR IN THE CENTER WITH MANY CONNECTIONS.
- SPECIALIZED AGENCIES ARE ON THE EDGES WITH FEW LINKS.



- THIS NETWORK CONFIRMS THAT NYC GOVERNMENT AGENCIES PRIMARILY HIRE AT THE EXPERIENCED LEVEL, WITH ONLY A FEW AGENCIES OFFERING A MIX OF ENTRY OR STUDENT-LEVEL POSITIONS. EXECUTIVE-LEVEL ROLES ARE SPARSE ACROSS THE BOARD.

RESULT

CONCLUSION

- NYC GOVERNMENT HIRING EMPHASIZES MID-LEVEL RECRUITMENT.
- A FEW AGENCIES DRIVE MULTI-CATEGORY HIRING, WHILE OTHERS ARE ROLE-SPECIFIC.
- SOCIAL NETWORK GRAPHS VALIDATED STRUCTURAL PATTERNS AND ROLE CONCENTRATION.

RECOMMENDATIONS:

- ENCOURAGE MORE ENTRY-LEVEL AND STUDENT ROLES TO SUPPORT WORKFORCE DEVELOPMENT.
- IMPROVE ROLE DIVERSITY IN SPECIALIZED AGENCIES TO ENHANCE FLEXIBILITY.
- USE SOCIAL NETWORK CLUSTERING TO IDENTIFY INTER-AGENCY COLLABORATION OPPORTUNITIES.
- INTEGRATE THIS MODEL FOR REAL-TIME HIRING ANALYSIS ACROSS GOVERNMENT DEPARTMENTS.

KEY FINDINGS

- EXPERIENCED (NON-MANAGER) ROLES DOMINATE NYC GOVERNMENT HIRING (71% OF POSTINGS).
- MOST JOB POSTINGS FALL IN THE $40,000–$80,000 SALARY RANGE.
- AGENCIES DIFFER IN SPECIALIZATION:
    - SOME FOCUS ON SPECIFIC JOB CATEGORIES (E.G., HEALTH, SOCIAL SERVICES).

NETWORK ANALYSIS REVEALED

- FUNCTIONAL SIMILARITY AMONG AGENCIES THROUGH SHARED JOB CATEGORIES.
- ENTRY-LEVEL AND EXECUTIVE ROLES ARE UNDERREPRESENTED.
- CLEAR AGENCY CLUSTERS BASED ON HIRING BEHAVIOR.

REFRENCES

1. NYC OPEN DATA. NYC JOBS - CURRENT JOB POSTINGS. HTTPS://DATA.CITYOFNEWYORK.
2. PYTHON NOTEBOOK JUPYTER FOR VISUALIZATION & ANALYSIS
3. PYTHON FOR CLUSTERING
4. EXCEL FOR FILTERING AND CLEANSING OF DATA
5. PYTHON FOR SOCIAL NETWORKS

ThankYou

BY Devansh Gupta





Gerente General